Web Engineering — Lab 2

Lab Task 1: Technical Article Webpage

Create an HTML webpage titled “Emerging Technologies in Computer Science.” Add a main heading and suitable subheadings. Write two paragraphs about artificial intelligence, cloud computing, and software engineering. Use different heading levels to organize the content, `<br>` to display information on separate lines, and `<hr>` to separate sections. Demonstrate bold, strong, italic, emphasized, highlighted, deleted, and inserted text. Use `<pre>` to display a short technical message while preserving spaces and line breaks.

Hints: `<h1>`–`<h6>`, `<p>`, `<br>`, `<hr>`, `<b>`, `<strong>`, `<i>`, `<em>`, `<mark>`, `<del>`, `<ins>`, `<pre>`.

Lab Task 2: Software Company Contact and Documentation Page

Create an HTML webpage for a fictional software company. Include the company name, a short mission statement using `<q>`, and abbreviations such as HTML and CSS using `<abbr>` with their full forms. Add the company's contact details using `<address>`. Demonstrate right-to-left text direction using `<bdo>`, display a sample program output using `<samp>`, and represent variables in a mathematical expression using `<var>`. Add HTML comments to identify the major sections of your source code.

Hints: `<q>`, `<abbr>`, `<address>`, `<bdo dir="rtl">`, `<samp>`, `<var>`, `<!-- comment -->`.

Lab Task 3: HTML Colors, Fonts, and Character Entities

Create a webpage titled “HTML Formatting and Character Entities.” Display a main heading and at least three sections. Demonstrate HTML colors using color names, RGB values, and hexadecimal values. Use the `<font>` tag to experiment with different font sizes, colors, and font faces, as taught in the lecture. Display the characters `<`, `>`, `&`, quotation marks, copyright, registered trademark, multiplication, and division using their appropriate HTML character entities. Separate the sections with horizontal lines.

Hints:

- Colors: `red`, `rgb(255,0,0)`, `#FF0000`
- Font tag: `<font color="blue" size="5" face="Arial">`
- Character entities: `<`, `>`, `&`, `"`, `'`, `©`, `®`, `×`, `÷`
- Other tags: `<h1>`, `<h2>`, `<p>`, `<hr>`